## Adding more layers vs lifting layer modification

| Model | FNO-CNN-4 layers | FNO-CNN-5 layers |
|---|---|---|
| Number of parameters | 4,281,954 | 5,359,330 |
| Training duration | 1321 | 1963.6 |
| Final loss | 5.0835E-08 | 4.8019E-08 |
| Average L2 error | 0.2792% | 0.2791% |

Above is the training results for FNO-CNN models with 4 and 5 layers. The number of FNO layers with the additional CNN branch is always 1 less than the total number of layers (so 3 and 4 layers).

The result of the regular FNO-CNN (3 total layers) is:

| FNO-CNN |
|---|
| 3,204,578 |
| 875.1 |
| **8.9862E-08** |
| **0.3929%** |

And the results of mixing FNO-CNN with different additional options for the lifting layer are shown here:

| Model | CNN input U-FNO | UNET input CNN | UNET input U-FNO | CNN input CNN |
|---|---|---|---|---|
| Number of parameters | 3,284,482 | 3,322,850 | 3,393,186 | 3,214,146 |
| Training duration | 1258 | 1766.1 | 2131.2 | 1164 |
| Final loss | 7.6996E-08 | 5.7729E-08 | 7.8146E-08 | 6.4734E-08 |
| Average L2 error | 0.3689% | 0.3203% | 0.3715% | 0.3449% |

It's then very obvious that modifying the lifting layer is a viable option as:

- Improve the final result at a small cost (only around 10,000 to 100,000 more parameters)
- Training time is acceptable, depending on configurations

With this, it's shown that UNET input into U-FNO is not a good configuration, as it takes longer to train, but the final result is not as good, comparing to FNO-CNN with even 5 layers.

CNN input CNN and UNET input CNN are the best options, especially the former.

- CNN input CNN final result is decent, but the training time and number of parameters are the smallest.
- UNET input CNN final result is very good for the number of parameters used, but the training time is quite long

On the other hand, if the number of parameters is not a constraint, going to 4 FNO-CNN layers is also not a bad choice. Then adding the CNN input modification (around 10,000 parameters) might be worthwhile.

However, going to 5 FNO-CNN layers is redundant for this problem size, as the result improvement is minimal but the extra cost is hefty.

## Training with non-continuous (Voronoi) material samples

| Model | FNO-Voronoi | U-FNO-Voronoi | FNO-CNN-Voronoi |
|---|---|---|---|
| Number of parameters | 3,149,090 | 3,274,914 | 3,204,578 |
| Training duration | 387.4 | 1029.3 | 1016.6 |
| Final loss | 2.2210E-06 | 1.1094E-07 | 1.6448E-07 |
| Average L2 error | 2.5718% | 0.5432% | 0.6696% |

Above is the training results with non-continuous material samples.

Here is the training results for continuous material samples:

| | Regular FNO | FNO with width = 33 | U-FNO | FNO-CNN |
|---|---|---|---|---|
| Number of parameters | **3,149,090** | 3,348,974 | 3,274,914 | 3,204,578 |
| Total training time (minutes) | **371.0** | 503.8 | 1343.0 | 875.1 |
| Final loss | 9.9875E-07 | 8.9477E-07 | 9.7017E-08 | **8.9862E-08** |
| Mean L2 error | 1.1864% | 1.1010% | 0.4185% | **0.3929%** |

It can be seen that, the 2 other methods perform very well, improving the results from 2.57% to 0.54% and 0.67%.  However, the U-FNO and FNO-CNN still suffer from the same ZSSR problem.




